Matrix Theory: The "Legal" Transformation Rules

In Engineering Math, we use **Elementary Row Operations (ERO)** to simplify matrices. These are the only three moves allowed to change a matrix without changing its **Rank**.

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1. The Three Legal Moves

You can perform these operations on any row to reach **Echelon Form** or **Normal Form**.

I. The Swap (Interchange)

You can swap any two rows. This is usually done to get a "1" or a non-zero number into the top-left corner (the pivot position).

- **Notation:** $R_i \leftrightarrow R_j$
- **Example:** Swap Row 1 and Row 2.

$$\begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} \quad \text{applying } R_1 \leftrightarrow R_2 \rightarrow \quad \begin{bmatrix} 1 & 2 \\ 0 & 5 \end{bmatrix}$$

II. The Scale (Multiplication)

You can multiply or divide an entire row by any non-zero constant (k).

- **Notation:** $R_i \rightarrow kR_i$
- **Example:** Divide Row 1 by 5 to get a leading 1.

$$\begin{bmatrix} 5 & 10 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

III. The Combine (Addition/Subtraction)

You can add or subtract a multiple of one row to another row. This is the primary tool for creating **zeros** below your pivots.

- **Notation:** $R_i \rightarrow R_i \pm (k)R_j$
- **Rule of Thumb:**
  - If signs are the **same**: Subtract ($R_2 - kR_1$)
  - If signs are **opposite**: Add ($R_2 + kR_1$)
- **Example:** Use Row 1 to create a zero in Row 2.

$$\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

2. Comparison: Determinants vs. Matrices

It is easy to get confused with 12th-grade rules. Here is the "How-To" difference:

Feature	In Determinants (12th)	In Matrices (Engineering)
Scalar k	Taken out of **one** row only.	Multiplies **every** element in the matrix.
Transformations	Done to find a **value**.	Done to find the **Rank** (Independence).
Goal	Make it easy to expand.	Create a "staircase" of zeros (Echelon Form).

3. The Golden Strategy: "The Pivot"

To avoid making mistakes, always follow this order:

1. **Get a "1"** in the diagonal position (e.g., A_{11}) using **Swap** or **Scale**.
2. **Kill everything below** that "1" using **Combine** ($R_2 \pm kR_1$, etc.).
3. **Move to the next diagonal** (A_{22}) and repeat.

> **Warning:** Never use a row that you are *currently changing* to change another row in the same step. Stick to using the "Pivot" row to change the others.